

The expenses on wear and tear of machines or capital equipments are known as depreciation allowances. It is necessary that a certain amount of money must be set aside to maintain the capital asset intact.

Sometimes tax is included in the price of the products which is the main source of government revenue. In order to estimate the accurate income of the businessmen, this tax should be deducted from the price of the products.

Various Concepts of National Income

The important concepts of national income in Macroeconomics are as under:

- (i) Gross National Product (GNP)
- (ii) Net National Product (NNP)
- (iii) Gross Domestic Product (GDP)
- (iv) National Income (NI)
- (v) Personal Income (PI)
- (vi) Disposable Personal Income (DPI)
- (vii) Per Capita Income (PCI)

Now these concepts of national income are highlighted:

1- Gross National Product (GNP)

The aggregate of all final goods and services produced in a country during a year is named as Gross National Product. It is the basic measure of economic activity. It includes agriculture products, industrial and mineral products and the production of other sectors of the economy. Similarly, the income of different people who provide services like doctors, teachers, lawyers, government officers etc. are also added. Moreover, the income/profit of services providing institutions like insurance companies, banks, shipping companies, means of transportation and communication are also included in gross national product. While adding up the goods and services, only the value of output currently provided should be considered. It excludes transaction in existing commodities i.e. old houses, other property, production of the previous year etc. Moreover, the following points should also be kept in view.

(i) Double counting is avoided: To determine gross national product, it is necessary that facts and figures should be accurate so that the desirable results can be obtained. To achieve this objective, we should count the value of only those goods which reach their final stage of production. For example, a shirt made of cotton is the final product, if we include the value of cotton as well as the value of shirt which is made of cotton, it will be double counting which should be avoided.

(ii) Counting of goods and services of personal use: In the measurement of gross national product, some items of personal use are not included. For example, if a personal car is sometimes used as taxi, it will raise national output but this is not counted in the estimation of national income. Similarly, personal house and services of house wife etc. are also example of such goods and services.

(iii) False facts and figures about incomes of the people and money received from hoarding, smuggling etc: Sometimes, people do not give exact information about their income. Some of the goods are smuggled or hoarded and are not counted towards national output. Hence, the estimates based on such an information are wrong. A sensitive estimate shows that 10% of the aggregate output is left uncounted because of illegal businesses and hiding of income

(iv) Counting of income received from abroad: Income received from other countries should also be counted in gross national product. Profits of all investors and incomes of individuals even working in other countries are included in gross national product. But the profits of foreign investors and incomes of foreign individuals are excluded because these incomes are remitted to the foreign countries.

In another way, if we estimate Gross National Product on the basis of total expenditure, the following components may be taken into consideration.

(v) Depreciation Allowances: It includes depreciation allowances.

(vi) At market prices: It values goods at market prices and not at factors costs, as indirect taxes are included in it.

(i) Private Consumption: Individuals of the country spend on food, clothing and shelter etc in a year.

ii) Public Consumption: Government spends on payments of salaries, allowances and other benefits etc during the year.

iii) Private Investment: Individuals investments out of their savings in private sector is called Gross private investment.

iv) Public Investment: Government expenditure made on public sector industries and other development projects.

(v) Foreign Investment: Foreign investors make investment in the country which is also a part of Gross National Product.

(vi) Exports minus Imports: (X-M) The difference between the value of exports and value of imports is also included in Gross National Product. Surplus balance of trade means that the value of exports is greater than the value of imports and the foreign exchange earning on export surplus becomes a part of Gross National Product.

Thus, Gross National Product = Consumption + Investment + Public Expenditure and Investment + (Exports – Imports)

$$GNP = C + I + G + (X - M)$$

2. Net National Product (NNP):

If the depreciation allowance or wear and tear expenses on capital goods/machines are deducted from Gross National Product, we get Net National Product.

Net National Product = Gross National Product – Depreciation Allowance

$$N.N.P. = G.N.P. - D.A.$$

Depreciation means the wear and tear expenses on machines, vehicles and other capital equipments. These expenses are included in Gross National Product but are not part of Net National Product. For example, a farmer purchases a tractor at a cost of Rs.3 lac. Say, the life span of the tractor is twenty years. If the farmer sets aside Rs.15 thousand per year for 20 years, for the replacement of the tractor, he will be able to purchase a new one when his present tractor is worn out in 20 years.

Depreciation is considered very important in the concept of national income because when we keep in mind the depreciation, capital remains intact and is not wasted.

3- Gross Domestic Product (GDP):

It is the market value of all goods and services produced during a year with the help of internal resources of the country. Different sectors of the economy need inputs to produce goods and services (output). This output is also called Gross Domestic Product. Neither this includes the income from external sources nor does the income which people residing abroad remit to the country.

Thus $\text{Gross Domestic Product} = \text{Gross National Product} - \text{Income from external sources}$

This can be explained with the help of facts and figures of Pakistan economy. According to 2011-12 data. Gross National Product of Pakistan at factor cost was Rs.6263187 million while income from external sources was Rs.234616 million so the Gross Domestic Product at factor cost was Rs.6028571 million.

4- National Income (NI) OR (Y):

We explained the concept of Gross National Product earlier. If the depreciation allowance or wear and tear expenses on capital goods/machines are deducted from Gross National Product, we get Net National Product.

The Net National Product includes indirect taxes such as; custom duty, excise duty, sales tax etc. If all these indirect taxes are deducted from Net National Product and subsidies are added, National Income is obtained. National Income is the sum of the payments of factors of production e.g. Rent, Wages, Interest and Profit. In another way, national income is also equal to the market value of goods and services.

$$\text{National Income} = \text{NNP} - \text{Indirect Taxes} + \text{Subsidies}$$

5- Personal Income (PI):

Personal Income means the income which individuals earn at their own during the period of a year. For example, a doctor earns Rs.5 hundred thousand in a year, it is his personal income. The following points should be remembered while computing personal income:

Transfer payments are included in personal income; any payment which people receive without doing labor, for example, Zakat, charity, scholarship, pension, gifts etc. These are not part of Gross National Product.

Direct taxes are also a part of personal income. So, personal income is an income before the payment of personal/direct taxes.

Personal Income does not include undistributed profits of joint stock companies.

$\text{Personal Income} = \text{National Income} + \text{transfer payments} + \text{direct taxes} - \text{undistributed profits of joint stock companies (corporate savings)}$

6- Disposable Personal Income:

Personal Income is the total income received by the individuals from all sources. All of the earned income cannot be spent by the individual. He has to pay some personal taxes and other payments such as: social security G.P. Fund B. Fund etc out of his personal income Suppose, any

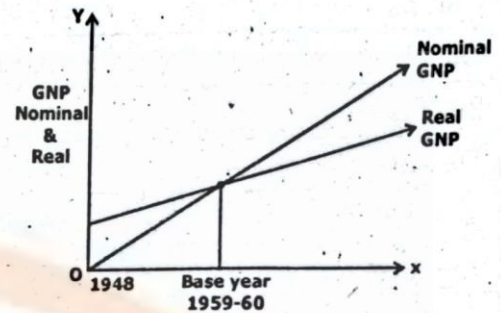
2010	Rs. 1000 billion	200	$\frac{1000}{200} \times 100 = 500$
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But if, the price remains the same as of base year i.e. 100 and value of output rises as known under, then the real G.N.P. will grow.

2010	Rs. 1000 billion	100	$\frac{1000}{100} \times 100 = 1000$
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Thus we can say that nominal (money) G.N.P. measure the level of output at prices prevailing in the period the output is produced 2010 in the schedule (Rs. 200) while, real GNP measures the output produced in any one period at the prices of some base year (2000) as shown in the above schedule.

The difference between the two may be shown in the following graph w.r.t Pakistan economy. The difference is due to change in prices.



It must be noted that the growth rate of the economy is the rate at which real GNP is increasing.

Methods of Measurement of National Income

The national income of a country can be measured by following methods.

1. Product Method (At market prices)
2. Income Method (Sum total of rewards of factors of production)
3. Expenditure Method

1- National Income at Market Prices (Product Method):

According to Dr. Marshall, "Aggregate of goods and services produced by utilizing the natural resources and capital in a country in a year, is known as National Income."

Thus by adding up the value of total goods and services produced, we get national income of the country. This includes the following:

Agriculture production like wheat, rice, sugarcane, cotton, vegetables, fruits etc.

Industrial production such as; textile, sugar, cement, machinery and other manufactured items.

Profits of services providing institutions such as: banks, insurance companies and shipping companies.

Income of the individuals who provide services like teachers, lawyers, doctors etc.

There are different units to measure the value of these goods and services. For example, units of length are meter and kilometer; the unit of weight is kilogram whereas, the unit of liquid is liter. The value of all these items is calculated on the basis of market prices. Similarly, the annual incomes of all the individuals providing services are also added. At the end, both the value of goods and services are added to attain the national income. This method to compute national

income is called "*National Income at Market Price*". It is further explained with the help of the following table.

Goods	Total Production	Market Price in rupees	Total Value in (crores Rs.)
Wheat	10 lac mounds	300 per mound	30
Rice	5 lac mounds	1000 per mound	50
Cotton	100000 bales	2000 per bale	20
Ghee	50,000 mounds	1000 per mound	5
Services			
Doctors	5000 individuals	100,000 per head	50
Teachers	10000 individuals	50,000 per head	50
Lawyers	5000 individuals	60,000 per head	30
Total National Income			235

Precautions:

While computing national income, it is very important to note that the data available is accurate so that: the estimates of income are not miscalculated. Because the development projects of the country largely depend upon true facts and figures of national income. Therefore, the following precautions should be kept in mind while computing national income:

i) Problem of double counting: In the measurement of national income, no item should be counted twice. For example, if we count the value of cotton in the national income as well as we count the value of cotton products like cotton yarn, cloths of cotton and garments, this is called double counting which results in over estimation of national income. To avoid double counting, either we should add the value of final goods and services, or we should count the additional value at each stage of production. For example, if we count the value of cotton in national income then cotton products should not be counted. However, the added value of the cotton products at each stage should be counted towards national income. Similarly, if the value of sugarcane is measured then the value of its product i.e. *sugar* should not be measured. In another way, the value of final products should be counted.

If, the value of the product is measured at various stages, the estimates of national income do not prove true, because the value of the product is counted again and again which is called double counting. This is further explained with the help of following table:

Stages of Production	Nature of the product	Value (in rupees)	Value added (in rupees)
1 st	Raw wood	10,000	10,000
2 nd	Pre-furniture	15,000	5000
3 rd	Furniture	25,000	10,000
Total		50,000	25,000

Above table shows that the actual value of furniture is Rs. 25,000 but due to double counting the value seems to be Rs. 50,000

ii) Hobbies/Recreational activities: The work which is done for the sake of hobbies or recreation is not counted in national income. For example, the value of vegetables grown around one's home as a part of one's hobby or rearing chickens or other pets which may be good source of meat for a family.

iii) Unpaid Services: Self provided services are also not a part of national income. For example, sewing one's own dress, knitting own pullover, preparing own food, polishing one's own shoes etc.

iv) Indirect taxes: The indirect taxes like sales tax, excise duty and custom duty should also be deducted from prices.

2- National Income at Factor's Cost (Income Method):

The second method to measure national income is the sum total of incomes of the individuals. As the national income is the income of a nation so, the incomes of all the persons of the country are added. In other words, national output is produced by the factors of production e.g land, labor, capital and organization. They receive their rewards/payments in return in the form of rent, wages, interest and profit. So if the factor payments are summed up, we arrive at national income. Now we explain briefly the rewards of the factors of production:

Rent: People earn money in the form of rent which includes rent of land, property, houses etc.

Wages: People also earn money in the form of wages of labourers, salaries of persons working in government or private offices, and other departments. Moreover, earnings of doctors, lawyers teachers are also their wages.

Interest: This item includes the interest on every type of loan. Interest on bonds issued by joint stock companies and payment on the use of capital is also included.

Profit: The earning of every type of business. Either the earning/profit is on individual business or on partnership basis, both of them are included in this head.

Thus

$$\text{National Income} = \text{Rent} + \text{Wages} + \text{Interest} + \text{Profit}$$

$$Y = R + w + i + \pi$$

Precautions:

The following precautions should be taken in this method:

i) Transfer payments: transfer payments should not be counted towards national income. Transfer payments means any payment which people receive without doing labor. For example, Zakat, charity, scholarship, pension, gifts etc.

ii) Illegal receipts: The money received from illegal sources such as; smuggling, bribery, robbery, theft etc should not be included in national income.

3- Expenditure Method:

According to this method if, we add every type of expenditure during a year, we get gross national product. If depreciation allowance and indirect taxes are subtracted from GNP and

subsidies are added, we get national income. The following expenditures are included to compute national income:

i) Private Consumption: The day to day spending of the individuals on food, clothing, Shelter etc. is added during a year.

ii) Public Consumption: Government also spends on defense, general administration, education, health, social welfare etc. during the year. These expenditures are also included in total expenditure.

iii) Private Investment: Individuals make investments out of their savings in private sector. For example, setting of new factories, constructing new houses, establishing new businesses etc. The sum of these investments is called Gross private investment.

iv) Public Investment: Government also invests on different projects such as; building new roads, new railway tracks, airports, power projects etc. Government expenditure in the public sector industries and development projects is added to get the national income.

According to Dr. Keynes:

National Income = Consumption + Savings or $Y = C + S$

National Income = Consumption + Investment or $Y = C + I$, where consumption plus investment shows total expenditure.

Thus: National Income = Total Expenditure

Circular Flow of National Income

As we have explained earlier that the national income of a country can be measured by three methods. The first two methods i.e. Product Method and Income Method are more important for computing national income. If sufficient precautions are observed while computing national income, each method will bring the same result. This is explained with the help of following diagram:



In this diagram, we have taken two sectors in the economy:

- (i) Households
- (ii) Business Firms

As far as households are concerned, they provide services of the factors of production such as; land, labor, capital and organization to the business firms whereas the business firms use

(c) Risk of tax evasion

(d) Decrease in national income

14- Marginal propensity to save is:

(a) Income - consumption

(b) Consumption = Savings

(c) Savings/ Income

(d) 1-MPC

Q-2. DESCRIPTIVE PART I (SHORT NOTES)**Give the short answer.**

(i) What is meant by national income?

Ans. The aggregate of all the goods and services which is produced in a country in a year is called national income. It includes agriculture products, industrial and mineral products and the income of different people who provide services.

(ii) Give the definition of national income given by Dr. Marshall.

Ans. "Aggregate of goods and services produced by utilizing the natural resources and capital in a country in a fixed period of time, which is generally a year, is known as National Income."

(iii) Define national income in three lines?

Ans. National Income is the value of flow of goods and services which is produced during a year in a country after deducting depreciation allowance and indirect taxes.

(iv) Define Gross national Product in three lines.

Ans. Gross National Product includes agriculture products, industrial and mineral products and the production of other sectors of the economy. It also includes the income received from abroad.

(v) Define Gross Domestic Product?

Ans. It is the market value of all the goods and services within a year by the internal resources of the country. It does not include the income received from external sources.

(vi) What is meant by Net National Product?

Ans. If the depreciation allowance is deducted from Gross National Product, we get Net National Product.

Net National Product = Gross National Product - Depreciation Allowance

(vii) Define Disposable Personal Income in three lines?

Ans. If direct taxes and social security payments etc are deducted from personal income, the remaining amount is disposable income which a person can spend. So $Y_d = Y - T$

(viii) Name the different methods used in the measurement of national income?

Ans: Product method or national income at market price

Income method or national income at factor cost

Expenditure method

(ix) What is meant by Per Capita Income?

Ans. When the national income is divided by the population of a country, Per capita income is obtained.

$$\text{Thus } \text{PerCapitaIncome} = \frac{\text{NationalIncome}}{\text{total Population}}$$

(x) Name the four factors upon which the national income of a country depends?

Ans. (i) Natural and human resources

(ii) Capital resources

(iii) Technology

(iv) Political stability

(xi) Differentiate between real and nominal income?

Ans. The total quantity of goods and services produced throughout the year is called real income. When the total goods and services are converted into money units, it is called nominal income. Nominal income fluctuates due to change in prices.

(xii) What is meant by marginal propensity to consume (MPC)?

Ans. MPC is the ratio of change in consumption to change in income.

$$\text{MPC} = \frac{\Delta C}{\Delta Y}$$

(xiii) What is meant by equilibrium level of national income?

Ans. Equilibrium level of national income means the national income is stable and there is no fluctuation in national income. According to Keynes national income is in equilibrium when:

$$\text{Aggregate demand} = \text{Aggregate Supply } Y = c + I$$

OR

$$\text{Savings} = \text{Investment} \quad s = I$$

(xiv) What is meant by Transfer Payments?

Ans. Any payment which people receive without doing labour, or carrying out any economic activity is called transfer payment for example, Zakat, charity, scholarship, pension, gifts etc.

(xv) What is meant by Depreciation Allowance?

Ans. Depreciation means the wear and tear expenses on machines, vehicles and other capital equipments. Because of depreciation capital remains intact and is not wasted.

(xvi) What is meant by Personal Income?

Ans. Personal Income means the income which individuals earn separately in a year. It includes transfer payments and direct taxes. Personal Income does not include undistributed profits of joint stock companies.

(xvii) How is national income measured by using total expenditure method?

Ans. According to this method, every type of expenditure is added to obtain national income of the country during a year such as consumption, investment and government expenditures.

(xviii) How double counting can be avoided?

Ans. To avoid double counting, either we should add the value of final goods and services, or we should count the additional value at each stage of production.

(xix) What is meant by Circular Flow of national income?

Ans. Households provide factor services to the business firms. In return, they receive rewards from business firms. Households spend their earned income on the purchase of goods and services. By doing so, the income spent goes back to the business firms for further production. This phenomenon is known as Circular Flow of National Income.

(xx) What is meant by marginal propensity to save (MPS)?

Ans. MPS is the ratio of change in saving to change in income.

$$\text{Marginal Propensity to Save} = \frac{\text{Change in Saving}}{\text{Change in Income}}$$

$$\text{MPS} = \frac{\Delta S}{\Delta Y}$$

PART II

1. Define national income. Discuss the importance of the study of national income.
2. Explain the different concepts of national income.
3. What is meant by national income? How it can be measured?
4. Explain the circular flow of national income with the help of a diagram.
5. What is meant by equilibrium level of national income?
6. Differentiate between marginal propensity to consume (MPC) and marginal propensity to save (MPS).



advance of loans; on the other hand the less the cash reserve ratio, the more the creation of credit.

6) Interaction of commercial banks:-

No single bank can create credit alone; it is rather the combined process of all banks. Because every bank retains a specific portion of deposit, and advances the loan. So a cooperation of all banks is required in the credit creation process.

7) Quantity of new deposits:-

The more the quantity of new deposits in the commercial banks, the more the capability of the banks to advance the loans. In this way, there will be more creation of money. But when the banks have less deposit, they will advance loans and create credit in a limited quantity.

8) Economic conditions:-

Ups and downs in the economic conditions of a country affect investment, growth and employment, so during recession or depression new investment comes to halt. So the demand for loans and the creation of credit slows down. Whereas during economic prosperity and expansion, there is rise in investment and employment. So there is more demand for loans and the creation of credit speeds up.

FUNCTIONS OF THE CENTRAL BANK

1) Note issuing agency:-

The primary function of the central bank is to issue currency notes in the country. In the absence of central bank every commercial bank could issue these notes; thus there used to be inflation in the economy which while no bank was prepared to accept the responsibility in this respect. So first of all the central bank was established in England, which was responsible for the issuance of currency notes and control of money supply in the economy. Nowadays, central bank in all the countries of the world is responsible for the issuance of currency notes. It also strengthens and manages currency system. There are two rules for the issuance of currency in the economy.

i) Fixed Fiduciary System:-

According to this rule, the limit to issue the notes is prescribed. This limit is set by the parliament of the legislative assembly. The central bank can print the currency notes against any reserve, but when they reach this limit, they have to keep gold, silver or foreign exchange against the excess amount as reserve. For example, if this limit is 5 millions pounds in a country, the notes up to 5 billion can be printed without any reserve. But if they have to print notes worth 10 million, gold, silver or foreign exchange are kept as reserve. This system is also prevalent in England, Norway and Japan. According to this system, the central bank cannot print surplus notes at its own, it has to seek the approval of the parliament, if it wants to change the limit. So there is no the risk of inflation but the drawback of this system is that it becomes inflexible, because the supply of money cannot be increased easily if the need arises.

ii) Proportional Reserve System:-

According to this system, a specific proportion is prescribed for the issuance of notes. Gold, silver or foreign exchange will be kept as reserve against the quantity of notes. For example, this proportion is 40% in America and Germany, 35% in France and 30% in Pakistan. It

means if the central bank of Pakistan wants to issue a hundred rupee note, it will have to keep Gold, Silver or foreign exchange of Rs.30 as reserve. In this system, the supply of money is flexible. The more quantity of Gold, Silver and foreign exchange in a country, the more power the central bank has to issue notes. But there is a risk of inflation in this system. Usually, the central bank issues notes according to these two rules. But Deviating from this rule, if the central bank issues money without keeping reserve, it is called deficit financing; this deficit financing causes inflation on one hand and destabilizes the value of money on the other.

2) Bank of the State:-

The central bank is called the bank of the state. All governments' transactions are done through this bank. This bank functions as the financial advisor of the state. It advises the government on financial matters. The government loans are received through this bank and government's payments are also made through central bank. In international trade, this bank provides foreign exchange. Export and import policy is also formed by its advice. If the quantity of foreign exchange is sufficient, easy import policy is adopted. Otherwise the commercial policy is tightened. In short, the supervision of all financial matters, assessment receipts and payments of the government are its responsibilities.

3) The bank of the banks:-

The central bank is called the bank of all commercial and cooperative banks. It controls the banking system of a country. All the banks in the country are under it. It supervises them completely. All the banks function according to the monetary policy of the central bank. This bank prescribes the timings of other banks. The policy of advancing loans and discount in other banks is also in accordance with its policy. Every commercial bank can open its branch in the country with its advice. It is also the supervisor of their investment. Commercial banks get loan when they are facing difficulties. As the commercial banks are registered with the central bank, the people deposit their money in these banks without any hesitation. Because they know that the central bank has complete control over these banks.

4) The custodian of metallic money and foreign exchange:-

The central bank is also the custodian of Metallic Money and foreign exchange in the country. The businessmen deposit all the foreign exchange they earn; and the central bank pays them in local currency. The foreign exchange sent by the people working abroad is also paid by the central bank in local currency. The gold and silver is also in its custody.

5) Lender of the last Resort:-

The central bank is the last resort of the commercial banks. If the commercial banks are facing any financial difficulty, they seek the help of the central bank and overcome their difficulty with its help. For example, if the commercial banks advance more loan or discount the bills more, and they have shortage of money to pay the creditors, they get loan from the central bank or get their bills re-discounted by the central bank and thus overcome their problem. If the central bank does not come to their rescue in need, it will be difficult for them to do business in the country and they lose people's trust.

6) Clearing house:-

The central bank also functions as the clearing house. The inter bank transactions are also made by central bank. All the representatives of commercial banks get together in the office of central bank to clear the transactions.

7) Controller of Money Market

The central bank of all countries performs many functions. But its main function is the issuance of currency and keeping its value stable. For specific purpose, the central bank adopts the methods to control the quantity of money which is called the monetary policy. Under this policy, it controls inflation and deflation. Being a controller of money market, it adopts the following methods to control credit money.

1) Quantitative Measures:

Quantitative measures are those methods which control the quantity of money directly. By these methods, the quantity of money can be increased or decreased. These methods are used to control both inflation and deflation. As in inflation, the quantity of money is surplus, so the measures are adopted under the monetary policy which decreases the circulation and quantity of money. Contrary to it, during deflation, the measures are adopted which increase the quantity and circulation of money. The important quantitative measures are stated as under:

i) Bank Rate Policy:

Bank rate policy means the rate of interest which the central bank prescribes when giving loans to commercial banks or discounting their bills of exchange. If there is inflation in the country, and the central bank wants to decrease the quantity of money, it increases bank rate. As a result the commercial banks also increase the rate of interest. Due to which the quantity of loan is decreased. Contrary to it, if there is recession and depressive situation owing to deflation, the bank increases the circulation of money. In this situation, it decreases the interest rate so that the commercial banks may get loans at low interest rates. Moreover, the rate of discounting the bills of exchange is also decreased. As a result, the circulation of money is increased and recessionary situation is controlled to some extent.

Conditions for Success:-

Bank rate policy to control inflation and deflation can be successful only when the following conditions are met:

All the institutions dealing with money like banks, agencies and different corporations should adopt the policy of the central bank completely. When the central bank increases interest rate, these institutions should also increase the interest rates and vice versa.

The economic system of the country should be elastic so that increase or decrease in credit money may increase or decrease rent, wages and supply of commodities. For this purpose it is necessary that the money market of the country should be organized and stable. As long as the different countries of the world were under the Gold Standard, gold coins were used as foreign exchange. The monetary policy played an effective role in safeguarding the reserves of gold and importing gold from other countries. When the central bank increased the interest rates, it gave incentive to other countries for investment, so they invested their money in the countries where the interest rates were high. Owing to it the gold reserves started to transfer to this country. Contrary to this, when the interest rates declined, the gold reserves started to transfer to the other country. Because, almost all the countries have said goodbye to gold standard, the bank rate policy is not much effective now-a-days.

For the success of Bank Rate Policy, it is essential that with the change in interest rate, the quantity of loans can be increased or decreased if the need arises. If the profit rate in the business is high, the increase in interest rate will not affect investment adversely. Because the people will be willing to take loan on high interest. So it will be difficult to control inflation. On



the other hand if the profit in the business is declining, the decrease in interest rate will not increase the quantity of loan, because the people will avoid investment and it will be difficult to control deflation.

ii) Open Market Operation:

The second strategy of the monetary policy of the central bank is open market operation through which the central bank buys and sells bonds and securities. For example if there is inflation in the country, and the central bank wants to decrease the circulation of money, the bank sells the securities and bonds in the open market. In this way the circulation of money decreases to some extent. Because the people buy bonds and securities and money goes to the bank. Contrary to it, if there is deflation, the central bank buys bonds and securities so the money reaches to the people and the circulation of money increases. In this way, through open market operation the quantity of money may be changed.

Conditions for Success:

The following conditions are essential for the success of open market operation.

It is necessary that when the central bank buys government securities, the cash reserves in the commercial bank must decrease.

When the cash reserves increase, the banks must give more loans and when the cash reserves decrease, the quantity of loan should also decrease.

But this condition may not be necessarily fulfilled because the demand for loan depends on industrial, economic and business conditions. If economic conditions are uncertain, the decrease in interest rate cannot increase the quantity of loans. If economic conditions are profitable, the quantity of loans cannot be decreased in this situation.

If there is increase or decrease in the cash reserve of the commercial banks which are lying with the central bank, the loans increase or decrease. It means the banks issue more loans if the cash reserves increase and they issue less loans if the cash reserves decrease.

The fourth condition is that there must not be any change in the velocity of circulation of credit money issued by the banks. Because the increase in the velocity of circulation of credit money cancels the effects of the policy of the decrease in the quantity of credit money. In this way decrease in the velocity of circulation of credit money will fail the policy of increase in the quantity of credit money.

There must be a broad and active market for the buying and selling of the short term and long term government securities.

Although, the above mentioned conditions affect the buying and selling of securities in the open market, this is worth mentioning that when the central bank sells or buys the securities, the power of the commercial banks to issue the loans increases or decreases. But in order to achieve this objective it is necessary that the money market of the country must be organized so that the securities may be bought and sold properly.

iii) Changes in Reserve Ratio:

As the central bank controls the scheduled commercial banks. The commercial banks can seek the help from the central bank when ever they want. In time of need, the central bank gives loans to the commercial banks and re-discounts their bills of exchange and saves them in the crises. Because of these services, the central bank keeps a specific rate of cash reserve out of their total deposits. For example if the rate is 5 % and if a bank has a deposit of worth 100 million rupees, the bank deposits 5 million rupees with the central bank and utilizes the remaining

amount itself. It is called the reserve ratio. If the central bank wants to control the inflation it increases this rate. For example if the central bank increases it to 6 %, now the bank has to deposit 6 million rupees instead of 5 million rupees. In this way, 1 million rupees more will be out of circulation.

Contrary to it, if there is deflation in the country, the central bank decreases this rate so that the banks may have more amount to grant loans. In this way the circulation of money increases and the intensity of deflation decreases to some extent.

Although this method is more effective than the other two methods, yet this method has also some limitations such as:-

If the commercial banks have a lot of deposits, they will be left with enough money to dilute the effect of the policy of the central bank by circulating money in spite of the increase in reserve ratio.

The commercial banks have comparatively easy access to cash deposits in the central bank. So, even if the central bank increases the rate, it makes no effect on the commercial banks.

The government usually prescribes low interest rates for the public debts, and the commercial banks are not willing to give loans on low interest rates. So the banks have excess money in the form of cash.

v) Credit Rationing:-

The central bank adopts this method to control inflation. If a bank does not abide by the policy of the central bank; sometimes, the central bank stops the credit of this bank. Owing to which the bank can not get loan more than the prescribed quantity, or its bills of exchange can not be re-discounted beyond the prescribed limit. In this situation, the banks are careful, because they fear that if they grant more loans or discount the bill of exchange, the central bank will not help them, and they will be in trouble. Thus the problem can be solved to a great extent.

The above mentioned methods to control the credit money are concerned with decreasing or increasing the quantity of money, so they are called Quantitative Measures. But it is necessary to explain that no above mentioned method can control inflation or deflation completely. In addition to them, there are other methods, which are discussed in the following lines:

2) Qualitative Measures:-

Qualitative measures do not directly affect the quantity of money, but indirectly they play an important role in the solution of the problem. they are stated as under:

Moral Persuasions: Some times the central bank morally persuades the commercial banks to adopt its policy and help to solve the problems in the country. We call this moral pressure, or moral persuasion.

Direct Action: If a commercial bank does not cooperate with the central bank and does not adopt its policy, the central bank threatens the commercial bank with direct action. The central bank refuses to give it loan or re-discount its bills of exchange. Moreover, central bank threatens it that it will strike off the name of bank from its list. So with the help of threats, the central bank forces the banks to help control the money according to its policy.

Publicity: Sometimes, the central bank controls inflation by publicity or advertisement. For example, the central bank of Pakistan has taken out millions of rupees from circulation through prize bonds. The people have bought prize bonds in the hope of winning prize and millions of rupees have been taken out of circulation and this money is kept with the central bank. The central bank persuades the people to save money through advertisements so that the people may

save money and the circulation of money may decrease. Publicity is a method to control the quantity of credit money.

Change in the Margin Requirements: According to this method, the bank grants the loans on different securities. The amount of loan is less than the value of security. The difference is called margin requirement. For example, if the value of a house is Rs.10 lacs and the bank grants the loan of Rs.8 lacs considering the price of house as Rs.8 lacs. So the margin is Rs.2 lacs. In order to control inflation, the bank increases the margin requirement so that less loans may be granted. Contrary to this, in the case of Deflation, the margin requirement is decreased.

Regulation of Consumer's Credit: According to this method, the consumer's buying power can be increased or decreased. This method affects the buying and selling of those things which are sold on installments. During inflation, buying of some goods is discouraged. For example, on the sale of these things, the first installments are heavy. Later the quantity of installments is decreased so that the consumers may not pay more money and the demand for this thing may not increase tremendously. During Deflation, the soft policy is adopted in order to increase the demand for consumers the payment of the installments is also made easy so that the expenditure is increased.

Meaning of Interest

In the Holy Quran, the word "Riba" often occurs for "Interest" (Usury). Therefore, it is necessary to throw light on the word "Riba" (Usury). Riba in Arabic language means "Excess" e.g., it is said that such and such person has gone to an excess with such and such person.

"Al-Riba" an Arabic word means "Excess and increase". Arabs would use this word "Al-Riba" for "Excess amount" in economic terms, the amount which a lender used get from the "Borrower" for the grace period or additional period or when the borrower would go beyond the fixed time to return the borrowed money. This increase in the principal amount lent to the lender was commonly called interest/ Usury (Riba). Before the revelation of the Holy Quran, all the forms of interest/ usury were called "Riba" by the Arabs. For example, whenever a person would sell anything to another person, he would fix a time for the payment. If the buyer could not make the payment on or before the time fixed by both of them, he (buyer) would ask extend the period of payment and for that purpose, he had to pay some additional amount (increase in the principal amount) which was called "Al-Riba" interest/ usury. Even if he could not make the payment, he would ask for further grace period and for that purpose further increase in the interest/ usury was fixed by the seller and the buyer with mutual consensus. This increase in the specific rate of interest/ usury was also called "Al-Riba".

There are two main types of Al-Riba usury/ interest.

1. "Riba Ansiya" (Interest/ usury on loan):-

This type of interest was commonly in practice during the period of "Ignorance". The lender or the seller would charge a specific rate of interest on the principal amount per month whereas the principal amount would remain payable and if the principal amount could not be paid by the buyer or borrower on the fixed day/ date and put it off under one pretext or the other, then the seller or the lender would increase the rate of interest or this amount.

2. "Riba-Fazal" (Net payment):-

In this type, the people would practice barter system i.e. they would exchange wheat with barley or similar other things of their need. In this method of Riba (usury) one person would owe something on the other for a certain period of time. If the same was returned no interest was

E-COMMERCE OR ELECTRONIC COMMERCE

It is defined as

"It is the application of communication and information sharing technologies among trading partners in the pursuit of business."

It can be divided into four parts.

1- Access to information:

Data stored is used for Productive and research purposes.

2- Inter-personal communication:

Increasing co-operation between two persons by sharing information.

3- Shopping Services:

Shopping is made through credit cards by internet.

4- Virtual experiences:

For education and trade purpose it is used to share information.

Instruments of E-Commerce

- (i) Electronic data interchange (EDI).
- (ii) BAR Codes.
- (iii) Electronic mail.
- (iv) Internet.

INFLATION

In simplistic terms the inflation is referred to increase in the price level. In other words, inflation is a situation where price hike of goods leads to devaluation of the currency. In economics the term "Inflation" means the demand of goods and services exceeds their supply which leads to price escalation. In this situation the supply of money increases but the quantum of goods and services remains unchanged. Thus, the balance between demand and supply of goods and services is disturbed and economy faces inflation.

According to A.P Lerner "Inflation is a situation where demand for goods and services exceeds their supply". Professor Henson wrote, "When prices of goods and services depicts increasing trend in relation to their production level, the situation is termed as inflation". In the words of Professor Crowther, "Inflation is a situation where value of currency falls and prices are rising". Gardener Ackley's opinion "when prices of goods are increasing persistently, then it is called inflation".

- 8- The inflation leads to overall resentment and uncertainty in the society. The labour demands more wages. Resultant, strikes, rallies and social unrest cause erosion of productive capacity.
- 9- The essential items of daily use consumes greater share of incomes, so the savings rate falls as a result of higher inflation.

The Problem of Inflation in Pakistan

Pakistan is a developing country and needs sizeable expenditure on development work. The process of development is painfully long and the results are visible after considerable time. The inflation has increased at an alarming pace in recent years. The inflation is persistently rising since 1950, however, the pace accelerated since 1970s. As a result of the hefty 131% devaluation in the early 1970s, the rupee lost its value from Rs.4.76 to a dollar to Rs.11 to a dollar. Our external debt rose almost 2.5 times in this period. The cost of imports rose substantially and we have to spend enormous amount of foreign exchange on imports. Industrial raw materials, crude oil and other commodities have witnessed substantial spike in the prices. Now Pakistan rupee lost its value substantially to almost Rs.95 to a dollar.

According to Economic Survey 2013-14, the CPI based inflation recorded as 8.7% percent. This is mainly driven by food inflation which is recorded 9.3% percent. This is mainly because of supply disruptions in wheat, vegetables, pulses, poultry, milk etc. The edible oil, wheat, tea, medicines witnessed an increase in world over.

Causes of Inflation in Pakistan

The inflation in Pakistan is mainly contributed by combination of factors like:

1. Increase in the Quantity of Money: The main cause of inflation in Pakistan is primarily the rising government expenditure need and the government has to demand more money from the State Bank. The currency in circulation in Pakistan has increased substantially from Rs.2 billion in 1950 to Rs.1.2 trillion in 2011-12. The prices have increased in response to this rise in base money.

2. Decrease in Production: During the last few years the quantity of money has grown much faster than growth in agricultural and industrial output. The agricultural sector has depicted a modest growth of around 3% in the last five years which is slightly above than the population growth. The unfavourable weather conditions beside salinity, drought, ill-planning, shortage of foreign exchange, lack of trained manpower, two wars of 1965 and 1971, lack of good management practices and entrepreneurial culture, are responsible for not recording sufficient growth in the productive capacity of the economy.

3. Population Pressure: One of the reasons for substantial spike in the price level is persistently higher population growth. The average growth of population in Pakistan was 1.5 percent between 1901 and 1951 which has risen to around 3% in 1972. The population growth in 2013-14 was 7.95% which is highest in the region. As a result of higher population growth, the resources are under pressure. Every year we have more people to feed, provide shelter and healthcare. The demand for goods is rising persistently and exerting pressure on prices. The total population of Pakistan is estimated at 188.02 million in 2014.

4. Devaluation: Pakistan has devalued its currency in 1955 and then in 1972. Consequently, the price of imported goods has increased substantially and external debt burden has gone up. The primary commodities became cheaper for foreigners and thus resulted in exports or smuggling, which in turn caused domestic shortage of these products. Pakistani rupee has depreciated substantially against dollar recently, which resulted in price hike.

5. Wage Increases: Different governments have increased minimum wages for labourers under labour policies but their productivity and efficiency has not increased. This led to cost push inflation in the country.

6. Increase in the Cost of Production: The prices of machinery and industrial raw material beside petroleum prices have gone up substantially at the international level in recent times. On the domestic front, the taxes and wages has gone up. The availability of skilled workers is not adequate which is also causing higher cost of production.

7. Development Expenditure: The government need enormous amount of money for their development program in the areas like irrigation, electricity, communication, information etc. These are often long-term programs and take many years for completion. During the period of project lot of money is spent but return are available on deferred basis. During this interim period the prices escalates and the government has to borrow additional amount to meet expenditure requirement on these programs. If adequate amounts are not available, the government asks the State Bank to print money which is inflationary in nature.

8. Deficit Financing: The government finances many projects through deficit financing over the years which has resulted in inflation. During the Second Five Years Plan, the government had arranged Rs.1.8 billion through deficit financing. During the Third Five Year Plan, the government amassed Rs.1.5 billion deficit financing for various projects. During the last three decades also the government has arranged deficit financing on regular basis which has resulted in inflation.

9. High Profit Taking Trend: Unfortunately, our industrialists and traders are more profit oriented and eager to utilize loose regulatory environment. Everybody wants to become rich in a flash and therefore, do not hesitate in using illicit means to grab more and more wealth. Extra market forces are persistently involved in hoarding, black marketing and other unfair means to get more wealth. These activities fuel inflationary pressures in the economy.

10. Unequal Distribution of Wealth: The distribution of wealth is very uneven in Pakistan. At the one hand, the rich is wasting millions of rupees on pomp and glory, and demonstration of power and wealth, but on the other hand one-fourth of the population is living below the poverty line. Almost half of these poor are chronic poor living at subsistent level. The rich is getting richer and spending million of rupees which are causing price hike of essential items and adding to overall increase in cost of living.

11. Oil and Petroleum Crisis: During the last few years, the crude oil prices are on the rise and they are fluctuating at very high levels. Pakistan is importing almost 60 percent of oil for its consumption. In 2011-12 Pakistan's import bill crossed \$ 10 billion mark and it was culprit of 43% of additional import bill. Pakistan has to spend considerable foreign exchange on petroleum imports. The rise in crude oil prices has resulted in overall hike in commodity prices in the international market. Pakistan has to import capital goods and industrial raw material from the international market. Thus, a general rise in prices of imports has caused inflation in Pakistan.

12. Social Evils: Many businesses in Pakistan are involved in extra market practices like hoarding of goods to create artificial shortages to earn exorbitant profits. They sell their goods in the black market at higher prices. Some businesses are even involved in smuggling of goods and they also export products to abroad illegally. Therefore, the artificial shortage of goods leads to higher prices. Our business men are also involved in adulteration, selling substandard quality products and cheating in weighing.

Measures to Control Inflation in Pakistan

The menace of inflation has become socially undesirable thing in Pakistan. To control inflation at a manageable level has become an important issue. The experts have suggested following measures to control the current higher inflation level:

- 1- Effective monetary policy is needed to control velocity and quantity of money at sustainable level. So that the currency in circulation may not cause inflation in the country. State Bank has taken several measures to control both through increase in bank rate, sale and purchase of bonds and securities etc.
- 2- Enhance the productive capacity of the economy through developing agriculture, trade, commerce, mining, communications and transportation sectors. The agriculture sector could be developed by introducing reforms, eliminating salinity and soil erosion, provision of quality seeds, insecticides and modernization of cultivation techniques. To promote industrial sector the private sector should be encouraged to invest, provision of industrial raw material conveniently at affordable rates, and tax incentives.
- 3- The population growth should be controlled effectively so that the demand is kept equal to supply. The imbalance of demand and supply is mainly responsible for price hike.
- 4- The exorbitant profit making should be checked so that businesses may not make money by using extra-market tendencies. The traders should feel responsibility for not making extraordinary profits at the expense of higher inflation for the masses.
- 5- The government should deal with hoarders, black marketers, and smugglers with iron hand and the society should be free of these unscrupulous elements. The general public will get goods at affordable rates.
- 6- The monetary policy should be used effectively to make income distribution more equitable. The rich should be taxed more and resources may be used for the welfare of the poor.
- 7- The relations between employee and employer may be made more cordial so that strikes and business closures are avoided.
- 8- There must be a complete ban on export of essential items to avoid shortages and make prices stable.
- 9- The goods which are not produced in adequate quantum may be imported to meet domestic consumption requirement to avert shortages of essential items. This will help in stabilizing their prices.
- 10- The administrative and non-development expenditure of the government should be controlled. The savings must be encouraged by taking necessary steps. The savings schemes like prize bonds, defense savings certificates, postal savings etc, should be

encouraged. The general public should be asked to follow austerity principles of our religion.

- 11- To keep inflation low we also need political stability which can help increasing productivity and better relationship between employee and employer. Investors take investment decisions under political stability. Only through higher investment, comprehensive planning, and administration could enhance production levels in the country.
- 12- In the short-run the government could impose control over prices of essential items like wheat flour, sugar, ghee, petrol, meat, cloth etc. The government could also control supply through provision of essential items through fair price shops or utility stores or even rationing of some items.

Measures taken by the Government

The government has taken various measures to mitigate the intensity of inflation burden on the general public at large. These measures include:

- 1- Ten top industrial units were nationalized in 1972 to establish the control of the government on industrial production. It was followed by nationalization of ghee units in the country to control supply of this essential commodity and stabilize prices of ghee.
- 2- State Bank of Pakistan has increased bank rate time and again to control currency in circulation and limit credit expansion. The bank rate was increased for the first time in 1959 when the bank rate was increased from 3% to 4%. This increase of bank rate continued until recently when the government decreased bank rate from 13% to 12% in June 2012. The State Bank also used variation in credit reserve ratio and statutory liquidity rate to control inflation.
- 3- The government has initiated various schemes to promote savings in the country like issuance of defence savings certificate, prize bonds scheme, postal life insurance, pension funds and behbood certificate etc.
- 4- The trade policy is being reviewed every year to incorporate the changing government policy and production patterns. The trade policy is directed to discourage imports by rationalization of tariffs and promote exports. The essential items of daily use are out rightly banned for imports to keep adequate quantity in the country. The government is also following exchange rate control policy in the country.
- 5- The government has also legislated punitive action against hoarders, black marketers, and smugglers to regulate supply in the country and discourage unscrupulous elements in the country.
- 6- The government has established price control committee in every district to monitor prices of items of daily use. There are various committees who raid violating traders to enforce strict price control in the country.
- 7- In addition to these price control committees at district level all district and provincial governments has special price control board with a mandate to keep a vigilant eye on the price variations. The federal government has also established "Agricultural Prices Commission" to monitor prices of four major crops.

PUBLIC FINANCE OF PAKISTAN

Introduction:

Under public Finance Revenue sources and expenditure items of Federal, provincial and local govt's are taken into consideration.

In Pakistan except Federal Government there are four provincial government's of Punjab, Sindh, N.W.F.P and Balochistan. Moreover, an autonomous govt. of AJK also functions under the supervision of Federal govt.

Federal and provincial government's have different fields authority areas to supervise, such as:

- 1- Federal government performs functions w.r.t Defence, currency, Transport communication, Information works, Foreign trade, Foreign affairs, petroleum, Minerals, power, energy resources etc.
- 2- Health, Education, population, social welfare, Industry, Agriculture, Housing, police, Public Administration matters are supervised by provincial governments.
- 3- Local Bodies, District and Tehsil councils, Municipal Corporations, Town committees can also change local rates to fulfill their requirements. Basic responsibility of Federal, Provincial and local governments is to provide food, Housing, Education, Health facilities Employment etc for the purpose of maximizing social and economic welfare of the masses. Budget for a Fiscal year is prepared for this purpose by each government.

BUDGET

Budget is a table or a statement which indicates Income estimates and Expenditure Heads of a govt. for a specific coming Fiscal year. In reality, Budget is an indicator of govt. policies regarding collection of income from different sources of Revenues and details of various Heads of expenditure during a certain time period (Coming Fiscal year)